CPG · MANUFACTURING PLANNING

Conceptual data model

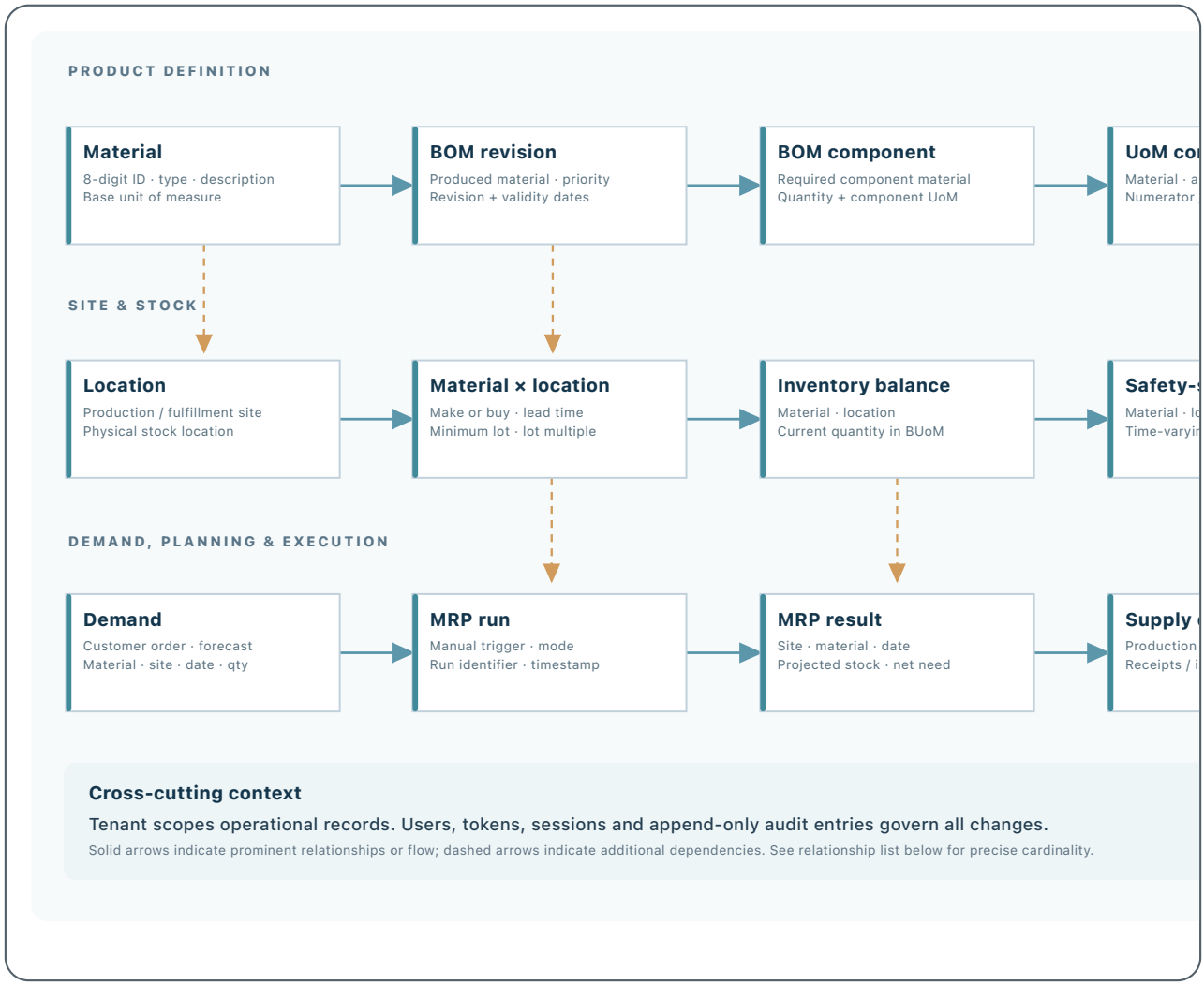
An entity map, business entity dictionary, relationship register and stock-flow diagram for the CPG MVP.

MVP functional baseline | 19 October 2026 | English source of truth

No stack or deployment decisions

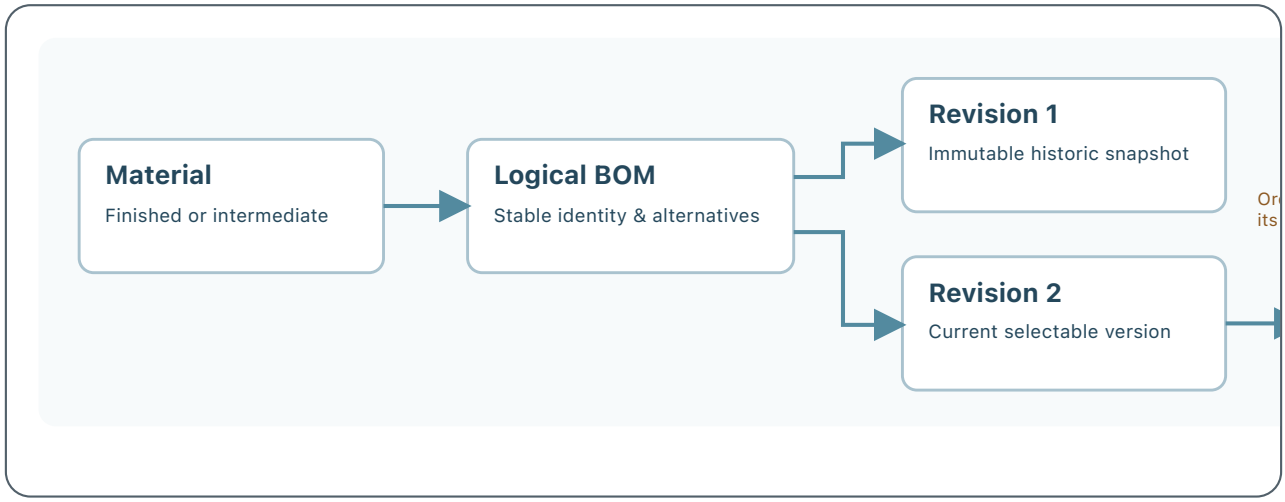
Visual conceptual entity model

This map shows business concepts and their most important relationships. It is deliberately independent of any future database or ORM.



Tenant scoping applies to all business objects. User, token, session and audit records span the operational model. Connectors indicate principal domain associations, not exact foreign-key cardinality. The relationship register below is authoritative for those relationships.

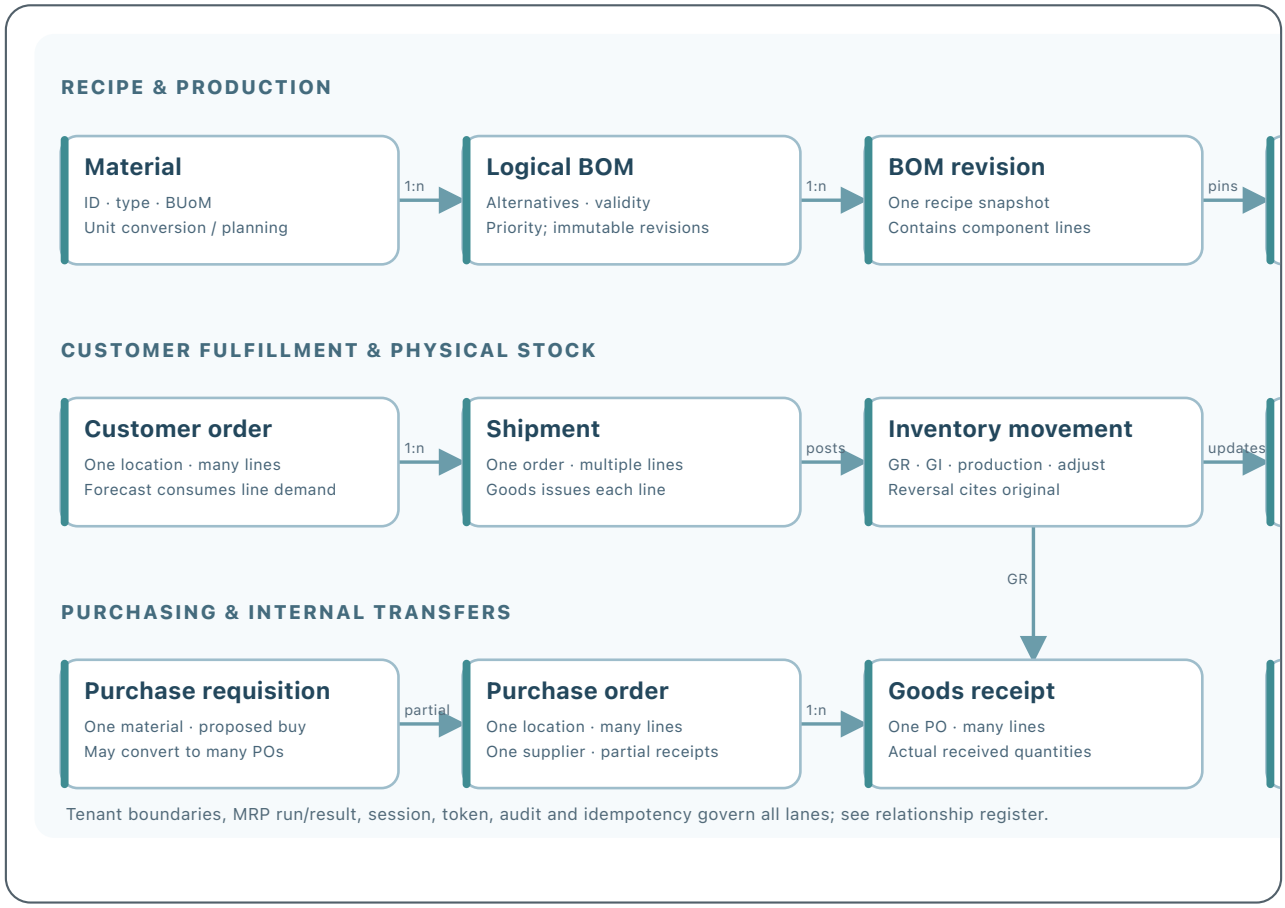
BOM version history and execution reference



Revisions are immutable and not treated as competing active BOMs. Changes to recipe, components, priority or validity create a revision; logical BOM activation/deactivation changes only its separately audited active flag. Existing production orders continue to use their pinned historic revision. The active

priority-overlap validation compares only currently selectable logical BOM alternatives for the same output material and overlapping validity periods.

Additional relationships: order lines, postings and transfers



Simplified many-to-one and transaction-flow connectors. Full cardinalities, including per-line receipt and issue reversal, are listed in the relationship register. This is a conceptual map, not an enforced physical database model.

Entity dictionary — complete MVP concepts

Logical objects, not a demand for a separate physical table per term. Lines and movement postings have their own audit identity to permit partial processing and full single-line reversals.

Business concept	Meaning and attributes	Kind
Material	Per-tenant dataset-generation eight-digit sequential immutable ID from 00000001, type raw/intermediate/finished, description, BUoM and active	**Master**

	flag. Prod IDs never reused; new Test generation restarts every business sequence. BUoM locks when any BOM revision, operational document or stock movement references the material.	
UoM / material conversion	Alternate-to-base conversion as positive numerator/denominator; BUoM stored immediately, original entered unit/factor retained for audit/history.	**Master**
Logical BOM	Stable alternative-recipe ID for output material, **independently audited active flag** and immutable revision history; toggling active status does not create a recipe revision.	**Master**
Immutable BOM revision	Version number, priority, inclusive validity from/to, output reference of one BUoM, complete component snapshot. Only changes to recipe, priority or validity create new revisions; unchanged upload/status toggle do not.	**Master**
BOM component	Positive raw/intermediate BUoM quantity per one output BUoM, historical entered UoM/factor. An input later deactivated stays in immutable history, but makes this alternative ineligible for new MRP order selection (explicit exception; fallback possible).	**Master**
Location	Physical production and stock site with active flag; prevent deactivation with nonzero physical/in-transit stock or open orders/transfers.	**Master**
Material-location parameters	Site Make/Buy, whole-calendar-day lead, minimum quantity and lot multiple. Raw Buy-only; intermediate/finished Make or Buy. Final output proposal must be rounded to minimum/multiple before testing whole BOM components.	**Master**
Safety stock target	Material × location × effective-from date target; persists until next target, zero default.	**Master**
Customer / supplier	Each has tenant-unique ID/name and active flag; historical open orders allowed after deactivation but no new orders.	**Master**
Physical inventory balance	Current nonnegative material × location BUoM balance from posted goods and adjustments.	**State**
In-transit inventory	Transfer-linked material in transit with nonnegative balance distinct from site stock.	**State**

Customer order header	Customer, one fulfillment location, lifecycle state, order identifier.	**Demand**
Customer order line	Material, ordered BUoM quantity/UoM, original requested delivery date, fulfilled GI quantity, abandoned quantity, line state; forecast-consumption linkage by date/material/site.	**Demand**
Forecast	Material/site/original date quantity and entered UoM (no customer). Orders consume it on their original due date; residual forecast from dates before today expires, whereas overdue unfulfilled firm demand is shifted to today.	**Demand**
Purchase requisition	One Buy material/site/required date/quantity, converted amount, status, optional source MRP run.	**Transaction**
Purchase order header	Supplier, one location, lifecycle state, optional direct creation without PR.	**Transaction**
Purchase order line	Material, quantity/UoM, planned date, receipts and optionally source requisition and converted quantity.	**Transaction**
Goods receipt header	One PO, time of actual posting, contains multiple GR lines; atomic all-lines posting.	**Transaction**
Goods receipt line posting	References one PO line, actual quantity (can over-receive, warn), material and location; may be reversed once in full.	**Transaction**
Outgoing shipment header	One customer order and location, planned date and status; may contain multiple lines.	**Transaction**
Shipment line / goods issue line posting	References customer order line, dispatched quantity and posting date; GI physically reduces stock. Entire issued line may be reversed once.	**Transaction**
Production order	One output material, site, completion target/date, computed start date, pinned immutable BOM revision, lifecycle status and optional MRP source.	**Transaction**
Production completion posting	Actual output (zero permitted), posted date, component-usage lines, planned/actual variance; each posting reversed as a whole once.	**Transaction**
Actual	Material, exact actual consumed BUoM quantity and	**Transaction**

component usage	expected BOM quantity, including added raw/intermediate inputs.	
Stock transfer	Single material, two distinct physical sites, planned dispatch and arrival with arrival not earlier. Planned quantity, partial postings and status. Completed only when all units actually received without loss; any write-off/abandonment resolves Closed Early once in-transit is zero.	**Transaction**
Transfer dispatch / receipt posting	Partial source goods issue to in-transit and partial destination goods receipt from in-transit; receipts never exceed dispatched quantity.	**Transaction**
In-transit write-off	Audited reasoned loss posting; full transfer closure requires zero in-transit; linked reverse permitted to restore transit before dispatch reversal.	**Transaction**
Inventory movement	Signed BUoM posting and reason/source (GR, GI, production, adjustment, transfer, reversal) with immutable date today.	**Ledger**
Inventory adjustment	Single material/location signed quantity derived from user input or XLS target current balance, reason required; opening balance and customer physical returns handled here.	**Ledger**
Reversal posting	Linked to original GR/GI line, adjustment, transfer posting or entire production completion; once, full amount, reason, current date; no negative balances.	**Ledger**
MRP run	User/time/tenant dataset generation, all locations, 90 daily inventory buckets, optional dated Beyond Horizon proposals, manual vs optional auto creation. Results and exceptions stored; hypothetical multilevel plan kept distinct from executable actions.	**Planning**
MRP daily result	Per material/site/day projected stock before/after proposed executable actions, firm/forecast/safety demand, tentative/dependable stock pipeline, shortage and chosen BOM. Weekly view sums flows but uses end-of-week balance and week's minimum/peak shortage, not a sum of balances.	**Planning**
MRP recommendation	Executable PR or fully feasible lot-rounded production order, or separately tagged hypothetical multilevel	**Planning**

/ exception	parent/intermediate actions never netted as supply. Include inactive-component / Missing Usable BOM, **At-Risk Existing Production** (open order with newly inactive pinned input), precise BOM shortages and overdue/late/beyond-horizon dates. At-risk output not dependable and never triggers automatic duplicate replacement.	
Tenant	Two isolated Test/Prod datasets, read-only flag; a Test reset regenerates a new dataset generation and restarts every Test business ID sequence at 00000001, preserving global user security.	**Governance**
User account / preferences	Exactly one built-in admin env password, normal users provisioned by admin, language/theme preference and active/inactive status.	**Governance**
Session / API token	Concurrent shared sessions; users may issue multiple named non-expiring bearer tokens only through explicitly confirmed interactive authenticated UI, never bearer API. Token list/revoke via UI or API; secret once, last use tracked, invalidate on account reset/deactivation.	**Governance**
API idempotency record	Account + tenant generation (or Global) + key, 24-hour request fingerprint/result for exposed mutating bearer APIs. Test API mutations require current dataset-generation precondition and old-generation requests fail even if old ID is reused; token issuance UI-only.	**Governance**
Tenant and Global audit events	Append-only before/after snapshots, account/session/token-channel identification; Test tenant audit cleared on Test reset, Prod/Global retained indefinitely.	**Governance**
Import dry run	Template, rows, errors/preview, source records' versions/fingerprint and Test generation, atomic commit. Stale-generation preview cannot commit. Material/BOM and partial physical-stock corrections.	**Workflow**

Conceptual relationship register

Readable cardinalities and invariants that accompany the diagrams. In particular, order lines and postings, not just headers, carry physical execution amounts.

Tenant generation → all business objects

Each entity belongs to Test or Prod (and the current Test dataset generation); never cross tenants or old Test generations even when a reset reuses IDs. Global accounts/tokens and Global audit remain outside tenant business data.

Material → logical BOM alternatives

One Make material can own many alternative recipes; raw never output, finished never component.

Logical BOM → immutable revisions

One logical BOM has many immutable recipe versions and an independent audited active flag. One currently selectable revision when active; toggling status does not create a revision, nor does unchanged import.

BOM revision → component lines

One revision has the complete historical recipe; each line references a raw/intermediate material. Quantity per one output BUoM.

Production order → immutable BOM revision

Many orders may pin a single version. Later modifications/deactivation never change the pinned version.

Material × location → planning parameters / inventory

One site-specific parameter set for Make/Buy, lead time and lot rounding; stock balance tracked separately.

Material × location × date → safety target

An effective-from series, each new value superseding previous target; default zero.

Customer → customer order → order line

One customer order has one fulfillment site, many material/date lines; each line has fulfilled and abandoned amounts.

Order line → shipment line → goods issue line

One order line may have many partial shipment and issue postings; one shipment belongs to one customer order.

Purchase requisition → PO lines

One single-material PR may convert into multiple purchase-order lines across POs/suppliers; tentative PR and converted PO supply must not double count.

Supplier → PO header → PO lines

One PO is for one supplier at one location and can contain many materials/dates.

PO line → goods receipt line

Many physical receipt postings per PO line; a GR document contains one or many lines from exactly one PO.

Goods receipt / issue line → reversal

One posting can have at most one linked complete reversal; other lines in a multi-line document are untouched.

Production order → completion → actual usage

One order may have many zero/partial/overproduced completion postings; each completion can record multiple consumed components and its own full reversal.

Transfer → partial dispatches / receipts

One material + source + destination.

Dispatch adds in-transit; receipt removes in-transit/adds destination. Partial permitted with nonnegative constraints.

Transfer → write-off / reversals

Write-off reduces in-transit for loss. Source issue reversal needs downstream

receipts and write-offs unwound first.

Posting → immutable movement ledger

Every physical or transit inventory change has signed BUoM movement, business date, type, source document and audit; posted transactions never silently edited.

Demand forecast ↔ customer order lines

Forecast is consumed by original order quantities at material/location/original due date, released for abandoned amount when closed early/cancelled; residual past-dated forecasts expire instead of being carried forward, while overdue firm demand is due today.

MRP run → daily results / action proposals

One 90-day all-site run per tenant generation; 90 dated inventory balances plus later Beyond Horizon proposals,

Open supply documents → MRP projections

PR tentative; PO and executable open production dependable, converted PR linked to PO once, transfers by stage.

rounded-BOM explanations, inactive-component and supply exceptions, separated executable actions/hypothetical dependency plans and created-order references.

Open production using newly inactive pinned component is separately At-Risk Existing Production: retain the order/commitment, exclude its output from dependable feasibility, block duplicate automatic replacements and show exception. Hypothetical multilevel actions are never dependable.

User → tokens/sessions/preferences

One account owns many named tokens issued by interactive UI only, simultaneous sessions and language/theme. Token list/revoke available via UI/API; password reset/deactivation invalidates credentials.

Tenant read-only → permitted writes

Read-only blocks business modifications across UI/API and MRP runs. Global admin and preference changes remain possible; protected Test reset temporarily enters read-only.

API request → idempotency result

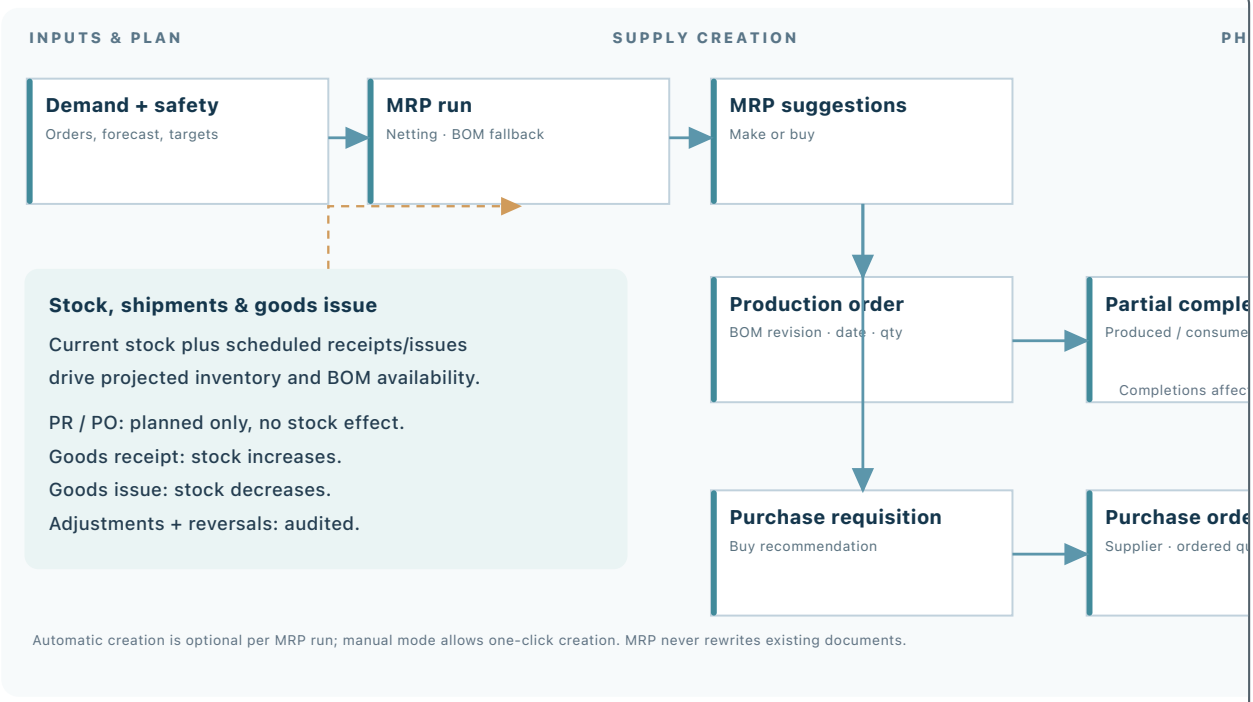
Every exposed bearer API mutation has account/scope/key/payload for 24h. Test business writes must carry current dataset-generation precondition and old-generation retries fail after reset even if IDs recur; same-generation identical retries replay the prior result. Token creation is UI-only.

Mutation → tenant or Global audit

Append-only snapshots with before/after and actor/channel; Global audit survives Test reset.

> **Active BOM uniqueness:** two distinct current selectable BOM alternatives for one produced material cannot share a priority across even partially overlapping validity ranges. Validate edits *and logical-BOM reactivation* without adding a recipe revision for status changes. Superseded revisions remain archived and exempt from the active-alternative collision check. Any direct or indirect component cycle is rejected with the full path shown.

Flow of documents and stock



MRP creates planned supply; the warehouse ledger only changes on posted GR/GI, production completion, transfer movement, reversal or adjustment. The figure shows the usual procurement chain, but direct purchase orders without requisitions are allowed.

Posting versus planning availability

Physical reality

Physical stock at a location changes on goods receipts, issues, production completions and corrections only. Transfer goods issue transfers quantity into distinct in-transit balance until destination receipt. Negative physical and transit quantities are prohibited. An outbound shipment or purchase requisition never directly changes stock; incoming procurement consists of PR, PO and actual GR, without an inbound-shipment entity.

MRP projection

Projected stock may be negative and may account for already scheduled receipts, tentative PRs (marked tentative), planned transfer arrivals and start-date BOM consumption. Existing PRs/POs and production orders persist through reruns. Partial physical postings reduce outstanding supply/need without double-counting.

Postings and reversals

Correct a posted GI/GR line, adjustment or entire production completion only with an opposite posting, full quantity once and mandatory reason, never direct edit. Production reversal subtracts finished output and restores actual component use; cannot proceed if finished stock insufficient. Transfer receipt reversal returns stock to in-transit.

Customer demand

Customer orders have lines dated individually; one outgoing shipment groups lines from just that one order. Fulfillment is posted with GI. Original customer quantities consume forecast until explicitly abandoned via early closure/cancellation; late open lines are due now for planning.

Model invariants (functional acceptance constraints)

- **Tenant boundaries:** Test and Prod have separate masters, ledger, orders, MRP, audit and ID sequences. Test reset starts a new dataset generation with all Test-local automatic business IDs reset to 00000001, rejecting stale API mutations and imports even where IDs repeat. Global users/tokens and Global audit survive.
- **Material keys:** sequential tenant-generation-local eight-digit zero-padded material IDs, immutable and never reused *within a generation*, but restart after Test reset. Raw Buy-only; intermediate/finished Make/Buy. Type locks after BOM or transaction reference; BUoM locks after *any* BOM revision, operational document or inventory movement reference.
- **Conversion:** entered alternate UoM is converted immediately via exact positive numerator/denominator, computed in decimal at sufficient intermediate precision and rounded half-up once to six BUoM decimals (reject invalid/rounded-zero). Keep original amount/unit/factor; stored BUoM remains authoritative. Planned component requirements and MRP supply round upward, and final output lots are rounded before checking full BOM feasibility. Never revalue historic BOM or ledger quantities.

- **BOM:** components raw/intermediate, output intermediate/finished; quantities per one output BUoM. On recipe/component/priority/validity change create immutable revision (not for identical import); audited logical-BOM active-flag toggles do not make a revision. All historic revisions remain. Reject any cycle. Prohibit equal priority for distinct current BOMs with overlapping valid dates; every edit validated.
- **Production:** each order pins exact immutable BOM revision. Partial completions, variances, added inputs, overproduction with warning and zero-output waste permitted. Completion posts actual input and output atomically; physical stock never goes negative.
- **Documents:** four states Open, Completed, Closed Early, Cancelled; partial based on quantity. Customer/PO can have many material lines, one location per order. Shipment and GR may have many lines but only one parent customer order/PO respectively. Production order and transfer single-material.
- **Fulfillment:** physical goods issue fulfills customer-order line and cannot exceed outstanding quantity; PO receipt can exceed ordered quantity with warning. Every outgoing shipment references one customer order. No customer returns module; corrections use inventory adjustment.
- **Forecast:** original customer line amount consumes same material/site/date forecast, GI fulfills firm need. Closed Early releases abandoned portion to residual forecast. Firm demand prioritized ahead of forecast and safety.
- **Stock ledger:** GR/GI/production/transfer/adjustment/reversal are actual immutable postings, each with document source and audit. Posting uses current business date only. Physical and in-transit never negative; projected stock can go negative to expose shortages.
- **Transfer:** distinct source/destination sites with arrival no earlier than dispatch; partial postings and in-transit not usable as physical stock. Cannot close with transit. Completed needs all planned quantity received without loss; any write-off or abandoned amount resolves Closed Early once transit is zero. Reversal of a write-off may automatically reopen transfer. Undo destination receipts/write-offs before source dispatch reversal.
- **Reversal:** whole line-level receipt/issue or whole individual production completion/adjustment, once, today, with mandatory reason and source link. Atomic stock checks, especially finished output must be available for production reversal. Existing historical postings unchanged.
- **MRP:** on-demand 90 daily buckets across tenant sites, plus dated Beyond Horizon proposals. Past residual forecast expires; overdue firm demand shifts to today.

Reruns net actual open supply, but production pinned to newly inactive inputs is reported At-Risk Existing Production and not treated as dependable or automatically replaced. Round output lots before full BOM test; new Make with inactive inputs gets fallback/exception. Hypothetical multilevel plans never become executable same-run supply. Start-date component projections and late/overdue warnings preserved.

- **Excel:** templates, all-errors dry run, dependency/duplicate validation, create/update preview, optimistic version and dataset-generation guard, all-or-nothing. BOM full-recipe import creates a new revision if changed. Opening/correction inventory imports specify target balances, not deltas; omitted material/location positions untouched.
- **Security:** exactly one built-in admin with required environment password; admin manages normal users. Tokens can be created only in the explicitly confirmed authenticated interactive UI, never bearer API; list/revoke via UI/API. Password reset/deactivation invalidate sessions/tokens. All users see both tenants; read-only tenant blocks business writes even by admin/API. UI/API same rules and optimistic locking. All exposed bearer API mutations require a 24h idempotency key; Test data writes also require the current dataset-generation precondition.
- **Audit:** tenant audit append-only, read-only for all, Test cleared on reset; Prod indefinite. Global admin audit indefinite and survives reset. No password or token secrets in either.
- **UX and boundaries:** accessible bilingual in-app help and tooltips, English/Polish language, dark/light modes, exact footer with two links. No separate returns/finance/capacity/batch/approval/work-center modules and no stack/deployment decisions.

Requirements boundary and resolution status

All former functional open questions have been given explicit approvals in the grilling session or resolved by minimal defaults recorded in the decision register. Those defaults clarify safe behavior of existing entities; they do not authorize additional modules.

Outside MVP: financial valuation, billing, refunds/returns workflow, batches or serial traceability, capacity-constrained production sequencing, hard reservations, supplier pricing/source restrictions, multiple administrators, SSO, approval workflows, working-day calendars, automated transfer creation, or a specific code stack/deployment platform.

For operational rules see main specification, for decisions see the decision register, and for the exhaustive validation baseline see the 535-scenario acceptance catalogue.

Caveman Planning Good! Made in Jozefoslaw with Tokens, Coffee and OpenMercato Skills during AITechLeaders training